

Finite-state residue traces for Hénon maps with polynomial coordinate weights

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Abstract

We construct a finite matrix that represents every global residue trace of a complex Hénon map with a fixed polynomial coordinate weight. For $H(x, y) = (p(x) - ay, x)$, where p is monic of degree $d \geq 2$ and $a \neq 0$, the weight at a point is $q(x)$ with $\deg q \leq m$. The cyclic fixed-point scheme at period n has length d^n , but the matrix constructed here has size $(B + 1)^2$, where $B = \lfloor \max(0, m - d + 1)(d + 1)/(d - 1)^2 \rfloor$ is independent of n . Its entries are integer-coefficient polynomials in the parameters. The proof expands a global residue into nearest-neighbour exponent flows and bounds each local flow by a column-stochastic propagation argument. The signed stability trace is $-\text{Tr}(W^n)$, and the associated generating function is $\det(I - zW)^{-1}$. All identities remain valid at degenerate parameters when the trace is defined by the full local-residue sum. For quadratic maps and quadratic coordinate weights, we obtain an explicit factorization of degree at most four. We also characterize when this trace sequence has an ordinary trace-class representative: exactly when W is nilpotent.

Keywords: Hénon maps; global residues; polynomial weights; transfer matrices; periodic points; dynamical determinants.

1 Introduction

Global sums over periodic points can be easier to compute than the individual points, but a finite computation at each period does not by itself give a finite representation valid at all periods. This distinction is explicit for a polynomial Hénon map. Its period- n orbit equations form a zero-dimensional complete intersection of length d^n , where d is the degree of the nonlinear polynomial. A separate quotient algebra therefore gives a finite answer for every n , while its dimension grows with n . The question considered here is whether one fixed matrix can represent all of these answers for a prescribed polynomial coordinate weight.

The observable is the all-complex signed stability sum. At a nondegenerate fixed point of H^n , its denominator is $\det(I - DH^n)$, with no absolute value. The numerator is the product of a single one-variable polynomial q along the first coordinates of the orbit. At a multiple fixed point, the pointwise quotient is undefined. We use the sum of local Grothendieck residues as the definition at such a parameter. This choice retains the complete fixed scheme and makes the parameter dependence algebraic.

Classical inputs. The global-residue algorithms of Cattani, Dickenstein, and Sturmfels [2, Theorem 2.3 and Lemma 4.2 of the author preprint] provide Laurent coefficient and Gröbner normal-form descriptions for a fixed polynomial system. Their pure-power initial-form hypothesis holds for the cyclic Hénon equations. The conversion from the cyclic Jacobian to the return derivative is the scalar cyclic Hill identity, contained in Molinari's block-tridiagonal determinant

formula [3, Eq. (1) of the author preprint v3]. Both ingredients are used here with their original ownership.

A preceding time-ordered Hénon calculation in the accompanying research notes [4, Lemma 4 and Theorem 5] already records the Hill relation for its family, the all-period unit-weight residue cancellation, a numerator given by the return-derivative trace, and a signed scheme-residue convention at degenerate points. The present result does not count any of these facts as a new contribution. Likewise, the Fredholm determinant argument used later is classical operator theory [1, Sec. 3 of the author manuscript].

Uniform finite-state result. The additional step is a local bound in the Laurent expansion. Each term has a left-going and a right-going nonnegative exponent at every cyclic position. Nonvanishing imposes a system of nearest-neighbour inequalities. A column-stochastic matrix redistributes these exponents, and the number of sites that can reach a fixed site in k steps is at most $2k + 1$. The resulting geometric series bounds every local exponent by a constant depending only on d and $\deg q$. After this bound, the residue sum is the trace of a finite edge-state matrix.

More precisely, for every degree bound m we construct one matrix W of size $(B + 1)^2$, with

$$B = \left\lfloor \frac{\max(0, m - d + 1)(d + 1)}{(d - 1)^2} \right\rfloor,$$

such that the signed residue traces satisfy $\tau_n = -\text{Tr}(W^n)$ for all $n \geq 1$. The same matrix works at every period, including periods one and two with their coincident neighbours. Its entries are polynomial in all source parameters, so specialization does not require a genericity assumption. The associated generating function is the reciprocal of $\det(I - zW)$.

The comparison made here is between this uniform construction and the specific fixed-system residue and Hill inputs just cited. No assertion of global priority follows from that comparison. The theorem concerns the stated coordinate-weight observation on all complex fixed points. It does not construct a natural transfer operator for real dynamics. Sections 2–4 give the complete proof. Section 5 treats quadratic weights and the ordinary trace-class boundary, and Section 6 records the scope and finite verification evidence.

2 Cyclic fixed schemes and the main theorem

Fix integers $d \geq 2$ and $m \geq 0$. Let

$$p(X) = X^d + \sum_{j=0}^{d-1} p_j X^j, \quad q(X) = \sum_{j=0}^m q_j X^j, \quad a \in \mathbb{C}^*. \quad (2.1)$$

All coefficients in (2.1) are arbitrary complex numbers. The actual degree of q may be smaller than m , and $q = 0$ is allowed. The map and its one-step weight are

$$H(x, y) = (p(x) - ay, x), \quad Q(x, y) = q(x) = q(\pi_1(x, y)). \quad (2.2)$$

Thus Q depends only on the first coordinate. The map is a polynomial automorphism because $a \neq 0$. All periods below refer to its original integer iteration.

For each $n \geq 1$, use indices in $\mathbb{Z}/n\mathbb{Z}$ and set

$$F_{n,i}(\mathbf{x}) = p(x_i) - x_{i+1} - ax_{i-1}, \quad 0 \leq i < n. \quad (2.3)$$

Incidences are added when neighbours coincide. In particular,

$$F_{1,0}(x_0) = p(x_0) - (1 + a)x_0, \quad (2.4)$$

$$F_{2,0}(x_0, x_1) = p(x_0) - (1 + a)x_1, \quad F_{2,1}(x_0, x_1) = p(x_1) - (1 + a)x_0. \quad (2.5)$$

The cyclic scheme Z_n cut out by (2.3) identifies with the fixed scheme of H^n by

$$(x_0, \dots, x_{n-1}) \mapsto (x_0, x_{n-1}). \quad (2.6)$$

Indeed, the recurrence $x_{i+1} = p(x_i) - ax_{i-1}$ gives $H^i(x_0, x_{n-1}) = (x_i, x_{i-1})$. Conversely, the first coordinates of a fixed orbit give (2.3). These are polynomial maps on the coordinate rings and are inverse by the same recurrence, so the statement also holds for nonreduced schemes. In period one, the fixed point is (x_0, x_0) .

Write

$$g_n(\mathbf{x}) = \prod_{i=0}^{n-1} q(x_i). \quad (2.7)$$

When all fixed points of H^n are nondegenerate, the pointwise signed stability trace is

$$\sum_{P \in \text{Fix}(H^n)} \frac{\prod_{i=0}^{n-1} q(\pi_1 H^i P)}{\det(I - DH^n(P))}. \quad (2.8)$$

Here $\text{Fix}(H^n)$ is the set of fixed points with their chosen starting positions. A primitive orbit of length dividing n is not identified with a single point in this sum.

Definition 2.1 (Trace at every parameter). For any parameters (2.1), define

$$\rho_n = \text{Res}_{F_n}(g_n), \quad \tau_n = -\rho_n, \quad (2.9)$$

where Res_{F_n} is the sum of the local Grothendieck residues at all finite common zeros of (2.3). The denominator order is $F_{n,0}, \dots, F_{n,n-1}$ and the differential order is $dx_0 \wedge \dots \wedge dx_{n-1}$. At a multiple zero the full local residue is used, rather than a quotient by a vanishing Jacobian or a deletion of the point. Define the formal generating function

$$D_{p,a,q}(z) = \exp \left(- \sum_{n \geq 1} \frac{\tau_n}{n} z^n \right). \quad (2.10)$$

The finite complete-intersection property needed for this definition is proved in Lemma 3.1. The minus sign in (2.9) is forced by the return derivative, as proved in Lemma 3.2. Consequently (2.9) equals (2.8) whenever all fixed points at that particular period are nondegenerate. To interpret every coefficient of (2.10) by pointwise sums, this nondegeneracy condition must hold for every n . The residue definition needs no such condition.

Set

$$M = \max(0, m - d + 1), \quad B = \left\lfloor \frac{M(d+1)}{(d-1)^2} \right\rfloor, \quad \mathcal{E} = \{0, \dots, B\}^2. \quad (2.11)$$

For nonnegative integers k, e , define a coefficient at infinity by

$$c_{k,e}(p, q) = [X^{-1}] \frac{q(X)X^k}{p(X)^{e+1}}. \quad (2.12)$$

The notation $[X^{-1}]$ refers to the Laurent expansion at infinity, not to an expansion at the origin. Let W be the matrix indexed by $\mathcal{E}$ with entries

$$W_{(r,\ell),(s,v)} = \binom{\ell+s}{\ell} a^\ell c_{r+v,\ell+s}(p, q). \quad (2.13)$$

Theorem 2.2 (Period-independent finite-state representation). *For every choice of (2.1) and every $n \geq 1$, the residue in Definition 2.1 is well defined and*

$$\rho_n = \text{Tr}(W^n), \quad \tau_n = -\text{Tr}(W^n), \quad D_{p,a,q}(z) = \frac{1}{\det(I - zW)}. \quad (2.14)$$

The same $(B+1)^2$ -dimensional matrix is used for all n . Each of its entries belongs to

$$\mathbb{Z}[p_0, \dots, p_{d-1}, a, q_0, \dots, q_m]. \quad (2.15)$$

Each coefficient identity in (2.14) is compatible with arbitrary parameter specialization within (2.1). The generating function identity holds formally and defines an analytic germ at $z = 0$.

The dimension bound is sufficient, not asserted to be minimal. In particular, some edge states may not participate in any closed path. The theorem is an algebraic representation of the specified trace sequence. It does not identify W , $-W$, or an infinite-dimensional extension with a natural Perron operator.

3 Residues, stability, and local coefficients

3.1 Finite schemes and the normal-form residue

Lemma 3.1. *For every $n \geq 1$ and every parameter choice, Z_n is a finite complete intersection of length d^n , with no common zero at infinity in the homogenized cyclic equations. For any polynomial $g(\mathbf{x})$,*

$$\text{Res}_{F_n}(g) = [x_0^{d-1} \cdots x_{n-1}^{d-1}] \text{NF}_{F_n}(g). \quad (3.1)$$

In particular, the residue of g_n is a polynomial in the parameters.

Proof. Choose any monomial order that first compares total degree. Since $d \geq 2$, the leading monomial of $F_{n,i}$ is x_i^d , including the two short-period cases (2.4)–(2.5). These leading monomials are pairwise relatively prime. The relatively prime leading-monomial criterion therefore shows that the $F_{n,i}$ form a Gröbner basis. The standard monomials are exactly

$$x_0^{j_0} \cdots x_{n-1}^{j_{n-1}}, \quad 0 \leq j_i < d.$$

They give a basis of the quotient algebra of size d^n . An ideal of finite colength generated by n polynomials in n variables is a zero-dimensional complete intersection. Its scheme length is the dimension just computed.

The top homogeneous equations are $x_i^d = 0$ for all i . Their only common affine zero is the origin, so they have no common projective zero. This verifies the no-infinity hypothesis for the global residue. The pure-power initial-form residue formula [2, Lemma 4.2 of the author preprint] gives (3.1). The formula includes multiplicities through local residues and requires no reducedness assumption.

All reductions use monic polynomials, so the division procedure never divides by a parameter or a discriminant. Applying (3.1) to g_n therefore gives a polynomial in the p_j , a , and q_j even at parameters where the fixed scheme is nonreduced. $\square$

3.2 The sign in the return denominator

At a cyclic point, put

$$v_i = p'(x_i), \quad A_i = \begin{pmatrix} v_i & -a \\ 1 & 0 \end{pmatrix}, \quad \mathcal{M}_n = A_{n-1} \cdots A_0. \quad (3.2)$$

The chronological order in (3.2) gives $\mathcal{M}_n = DH^n(x_0, x_{n-1})$.

Lemma 3.2 (Cyclic Hill sign). *Let $J_n = DF_n$ be the cyclic Jacobian, with all coincident incidences added. Then*

$$\det J_n = \text{Tr}(\mathcal{M}_n) - 1 - a^n = -\det(I - \mathcal{M}_n) \quad (3.3)$$

for every $n \geq 1$. Hence the residue trace (2.9) agrees with the pointwise sum (2.8) on the nondegenerate locus.

Proof. For $n \geq 3$, the cyclic matrix J_n has diagonal v_i , right-neighbour entries -1 , and left-neighbour entries $-a$. In the scalar cyclic formula of Molinari [3, Eq. (1) of the author preprint v3], substitute these entries. The two corner products contribute

$$(-1)^{n+1}((-1)^n + (-a)^n) = -1 - a^n,$$

and the transfer-product term is $\text{Tr}(\mathcal{M}_n)$. Its nonzero off-diagonal assumptions hold since $a \neq 0$. This gives the first equality in (3.3). Also $\det A_i = a$, so $\det \mathcal{M}_n = a^n$. The identity $\det(I - M) = 1 - \text{Tr } M + \det M$ for a two-by-two matrix proves the second equality.

For $n = 1$, direct differentiation gives $\det J_1 = v_0 - 1 - a$, while $\text{Tr } \mathcal{M}_1 = v_0$. For $n = 2$, the neighbours coincide and

$$J_2 = \begin{pmatrix} v_0 & -(1+a) \\ -(1+a) & v_1 \end{pmatrix}, \quad \text{Tr}(A_1 A_0) = v_0 v_1 - 2a.$$

Thus

$$\det J_2 = v_0 v_1 - (1+a)^2 = \text{Tr}(A_1 A_0) - 1 - a^2.$$

The sign is therefore independent of n .

At a simple common zero, the local residue of g is $g/\det J_n$. By (3.3), its negative equals $g/\det(I - DH^n)$. Summing over the fixed points and using (2.6) proves the final assertion. $\square$

3.3 A local exponent expansion

The next formula separates the single-coordinate residue coefficients from the nearest-neighbour combinatorics. Choose a radius R large enough that

$$|p(X)| > (1 + |a|)R \quad \text{when } |X| = R. \quad (3.4)$$

Such a choice exists since $d \geq 2$. It depends on p, a , but not on n .

Lemma 3.3. *For each $n \geq 1$, the global residue ρ_n is the convergent sum*

$$\rho_n = \sum_{L_0, R_0, \dots, L_{n-1}, R_{n-1} \geq 0} \prod_{i=0}^{n-1} \binom{L_i + R_i}{L_i} a^{L_i} c_{R_{i-1} + L_{i+1}, L_i + R_i}(p, q). \quad (3.5)$$

If a product in this sum is nonzero, then with $e_i = L_i + R_i$ it satisfies

$$de_i \leq m - d + 1 + R_{i-1} + L_{i+1} \quad (0 \leq i < n). \quad (3.6)$$

Proof. The global coefficient formula for a system with these pure-power initial forms [2, Theorem 2.3 of the author preprint] identifies its residue with the coefficient of $x_0^{-1} \cdots x_{n-1}^{-1}$ in the expansion at infinity. Equivalently, on a sufficiently large product torus it is

$$\rho_n = \frac{1}{(2\pi i)^n} \int_{|x_0|=\dots=|x_{n-1}|=R} \frac{g_n(\mathbf{x}) dx_0 \cdots dx_{n-1}}{\prod_{i=0}^{n-1} F_{n,i}(\mathbf{x})}, \quad (3.7)$$

where each circle has positive orientation and the factor order is as in Definition 2.1. In this pure-power setting the large-torus formula is the coefficient formula just cited. The lower degree terms may be expanded as convergent geometric series at large R , and regrouping the terms involving only x_i gives

$$\frac{1}{F_{n,i}} = \sum_{L_i, R_i \geq 0} \binom{L_i + R_i}{L_i} \frac{a^{L_i} x_{i-1}^{L_i} x_{i+1}^{R_i}}{p(x_i)^{L_i + R_i + 1}}. \quad (3.8)$$

Condition (3.4) makes this expansion uniformly absolutely convergent on the integration torus. For each fixed n , its finite product can therefore be integrated term by term. Uniformity of the whole product norm in n is not needed.

The variable x_i receives R_{i-1} powers from its left neighbour and L_{i+1} powers from its right neighbour. Its one-variable integral is consequently $c_{R_{i-1}+L_{i+1}, L_i+R_i}$. Multiplying the local factors proves (3.5).

For $n = 1$, both neighbouring factors use the same variable. Summing over $L_0 + R_0 = e$ produces the coefficient $\sum_{L_0=0}^e \binom{e}{L_0} a^{L_0} = (1+a)^e$, as required by (2.4). For $n = 2$, the left and right neighbours again coincide, but their exponents add in the same variable. Formula (3.5) therefore retains the correct two incidences without treating them as different coordinates.

Finally, if $c_{k,e} \neq 0$, the highest possible power at infinity in (2.12) cannot be below X^{-1} . Hence

$$m + k - d(e + 1) \geq -1. \quad (3.9)$$

Apply this necessary condition to each local factor of a nonzero product to obtain (3.6). No sufficiency claim is made for this degree condition. $\square$

4 A uniform flow bound and finite-state closure

A total-degree estimate would only bound the sum of the exponents by a quantity proportional to n . The finite-state construction instead requires a bound at each position. The one-dimensional geometry of the cyclic index set supplies that bound.

Lemma 4.1 (Period-independent local flow bound). *Let $n \geq 1$, $d \geq 2$, and $\mu \in \mathbb{Z}$. Suppose nonnegative integers L_i, R_i , indexed cyclically, satisfy*

$$d(L_i + R_i) \leq \mu + R_{i-1} + L_{i+1} \quad (0 \leq i < n). \quad (4.1)$$

If $\mu < 0$, no such configuration exists. If $\mu = 0$, every exponent is zero. If $\mu > 0$, then for every i ,

$$L_i + R_i \leq \frac{\mu(d+1)}{(d-1)^2}. \quad (4.2)$$

Proof. Set $e_i = L_i + R_i$. Summing (4.1) over the cyclic index set gives

$$(d-1) \sum_i e_i \leq n\mu. \quad (4.3)$$

This is impossible for $\mu < 0$. For $\mu = 0$, nonnegativity forces all e_i , and hence all L_i, R_i , to vanish.

Assume now that $\mu > 0$ and fix one configuration. Construct an $n \times n$ nonnegative matrix P column by column. If $e_j > 0$, put L_j/e_j in row $j-1$ of column j and R_j/e_j in row $j+1$ of that column. Add the two contributions if the rows coincide. If $e_j = 0$, put 1 in row $j+1$ of column j . All unspecified entries are zero. Each column has sum one, and the zero-flow convention does not affect Pe . Specifically,

$$(Pe)_i = R_{i-1} + L_{i+1}. \quad (4.4)$$

The inequalities become $d\mathbf{e} \leq \mu\mathbf{1} + P\mathbf{e}$, componentwise. Because P is nonnegative, iterating this inequality K times gives

$$\mathbf{e} \leq \frac{\mu}{d} \sum_{k=0}^{K-1} d^{-k} P^k \mathbf{1} + d^{-K} P^K \mathbf{e}. \quad (4.5)$$

Every power of P is column-stochastic. Thus each of its entries lies between zero and one, and

$$\|P^K \mathbf{e}\|_\infty \leq \|\mathbf{e}\|_1. \quad (4.6)$$

Here n and the configuration are fixed. Since $d > 1$, the last term in (4.5) tends to zero as $K \rightarrow \infty$. There is no need to bound $\|\mathbf{e}\|_1$ uniformly in n at this stage.

To control the first term, fix a row i . The support of P allows only steps between adjacent cyclic positions. If $(P^k)_{ij} \neq 0$, then the cyclic distance between i and j is at most k . There are at most $\min(n, 2k+1)$ such starting columns j . Using the entrywise bound from column-stochasticity gives the row-sum estimate

$$(P^k \mathbf{1})_i = \sum_j (P^k)_{ij} \leq 2k + 1. \quad (4.7)$$

In particular, the proof does not mistake a column-stochastic matrix for a row-stochastic one. The same estimate is valid for $n = 1, 2$, where the adjacent positions may coincide.

Take $K \rightarrow \infty$ in (4.5) and use (4.7). Then

$$\begin{aligned} e_i &\leq \frac{\mu}{d} \sum_{k \geq 0} \frac{2k+1}{d^k} \\ &= \frac{\mu}{d} \left(\frac{d}{d-1} + \frac{2d}{(d-1)^2} \right) = \frac{\mu(d+1)}{(d-1)^2}, \end{aligned} \quad (4.8)$$

which proves (4.2). $\square$

The auxiliary matrix P depends on the exponent configuration. It is only a proof device, not a Markov model of the Hénon dynamics. Its entries are nonnegative even when p, a, q are complex. The quadratic case $d = 2$ is covered without a denominator $d - 2$.

Proof of Theorem 2.2. Lemma 3.1 gives the finite scheme and defines the residue for all parameters. By Lemma 3.3, each nonzero product in (3.5) satisfies (4.1) with $\mu = m - d + 1$. Lemma 4.1 shows that either all products vanish, only the zero configuration contributes, or

$$0 \leq L_i, R_i \leq L_i + R_i \leq B \quad (4.9)$$

for every i . The integer part in (2.11) is justified because $L_i + R_i$ is an integer. Hence every nonzero term of the convergent residue sum lies in a single finite box independent of n .

Associate to a cyclic configuration the edge states

$$E_i = (R_{i-1}, L_i), \quad E_{i+1} = (R_i, L_{i+1}). \quad (4.10)$$

If $E_i = (r, \ell)$ and $E_{i+1} = (s, v)$, the i th local factor in (3.5) is

$$\binom{\ell + s}{\ell} a^\ell c_{r+v, \ell+s} = W_{E_i, E_{i+1}}.$$

Conversely, any labelled closed sequence of edge states determines the L_i as its second coordinates and the R_i as the first coordinates of the following states. This is inverse to (4.10), including for $n = 1, 2$. There is no quotient by cyclic rotation. Consequently

$$\begin{aligned} \rho_n &= \sum_{E_0, \dots, E_{n-1} \in \mathcal{E}} W_{E_0, E_1} W_{E_1, E_2} \cdots W_{E_{n-1}, E_0} \\ &= \text{Tr}(W^n). \end{aligned} \quad (4.11)$$

Additional states or configurations inside the box with zero product do not affect this equality. When $m - d + 1 \leq 0$, $B = 0$ and the same identity follows from the vanishing or the unique zero configuration already established.

For the coefficient ring, set $T = 1/X$. The reversed monic polynomial is

$$\tilde{p}(T) = T^d p(1/T) = 1 + p_{d-1}T + \cdots + p_0T^d.$$

Every coefficient of $\tilde{p}(T)^{-(e+1)}$ is an integer-coefficient polynomial in the p_j . Indeed, expand $(1 + U)^{-(e+1)}$ in powers of $U = p_{d-1}T + \cdots + p_0T^d$, whose constant term is zero. A given T -coefficient

receives only finitely many terms, each with an integer binomial coefficient. Formula (2.12) therefore gives an integer-coefficient polynomial in the p_j, q_j . The extra binomial factor and a^ℓ in (2.13) prove (2.15).

Definition 2.1 now gives $\tau_n = -\text{Tr}(W^n)$. The finite-dimensional identity

$$-\log \det(I - zW) = \sum_{n \geq 1} \frac{\text{Tr}(W^n)}{n} z^n \quad (4.12)$$

holds as a formal series. Substitution into (2.10) yields the reciprocal determinant in (2.14). Both the residue coefficients and the entries of W are polynomials, so the coefficient identities specialize at any parameter, including a drop in the actual degree of q . Finally $\det(I - zW)$ has constant term one, hence its reciprocal is analytic near $z = 0$. $\square$

Remark 4.2 (The low-degree threshold). For $m < d-1$, the negative-surplus case gives $\rho_n = \tau_n = 0$ for all n . For $m = d-1$, only $L_i = R_i = 0$ contributes and $c_{0,0} = q_{d-1}$. Thus

$$\rho_n = q_{d-1}^n, \quad \tau_n = -q_{d-1}^n, \quad D_{p,a,q}(z) = \frac{1}{1 - q_{d-1}z}.$$

The subcritical vanishing is the classical Euler–Jacobi degree phenomenon [2, Corollary 1.18 of the author preprint]. It is not the additional uniform-flow assertion needed for larger m .

5 Quadratic weights and ordinary trace representatives

5.1 An explicit factorization for all quadratic weights

Let

$$p(X) = X^2 + bX + c, \quad q(X) = uX^2 + vX + w, \quad a \in \mathbb{C}^*. \quad (5.1)$$

We use the degree bound $m = 2$, so $B = 3$ even if $u = 0$.

Proposition 5.1. *For all parameters (5.1),*

$$D_{p,a,q}(z)^{-1} = (1 - uz)(1 - auz)(1 - (v - bu)z - au^2z^2). \quad (5.2)$$

In particular, this polynomial has degree at most four and is independent of c and w .

Proof. The matrix (2.13) initially has the sixteen states $\{0, 1, 2, 3\}^2$. From (3.9), an entry from (r, ℓ) to (s, v) can be nonzero only when

$$r + v \geq 2(\ell + s) - 1. \quad (5.3)$$

A state with no incoming edge or no outgoing edge cannot occur in a closed walk. Deleting it preserves all power traces and $\det(I - zW)$: in the latter determinant its row or column has only the diagonal entry 1 after the corresponding zero row or column of W is used.

Using (5.3), delete the following groups in order, always in the current graph:

$$\begin{aligned} &\{(0, 3), (1, 3), (3, 0), (3, 1)\}, \\ &\{(0, 2), (2, 3), (2, 0), (3, 2)\}, \\ &\{(1, 2), (2, 1), (3, 3)\}. \end{aligned}$$

In each group every listed state has no incoming or no outgoing edge before that group is removed. This follows directly from (5.3). The remaining five states, in order, are $(0, 0), (0, 1), (1, 0), (1, 1), (2, 2)$.

Put

$$\begin{aligned} A &= v - bu, & C &= b^2u - bv - cu + w, \\ E &= -b^3u + b^2v + 2bcu - bw - cv. \end{aligned} \quad (5.4)$$

The Laurent coefficients in (2.12) give the following remaining block directly:

$$W_* = \begin{pmatrix} A & C & 0 & u & 0 \\ 0 & au & 0 & 0 & 0 \\ C & E & u & v - 2bu & u \\ au & a(v - 2bu) & 0 & 0 & 0 \\ 0 & a^2u & 0 & 0 & 0 \end{pmatrix}. \quad (5.5)$$

For example, $c_{0,0} = A$, $c_{1,0} = C$, $c_{2,0} = E$, $c_{1,1} = u$, and $c_{2,1} = v - 2bu$, which account for the displayed entries together with their binomial and a factors.

The state $(2, 2)$ is still a noncyclic bridge: its only possible incoming edge is from $(1, 0)$ and its only possible outgoing edge is to $(0, 1)$, from which there is no return. Its presence affects neither a power trace nor $\det(I - zW_*)$. Among the other states, the only possible nontrivial cyclic blocks are

$$[u], \quad [au], \quad \begin{pmatrix} A & u \\ au & 0 \end{pmatrix}. \quad (5.6)$$

After a simultaneous reordering of rows and columns, the matrix is block triangular with these blocks and the zero bridge block. Therefore

$$\det(I - zW) = \det(I - zW_*) = (1 - uz)(1 - auz)(1 - Az - au^2z^2).$$

Theorem 2.2 proves (5.2). All deletions and coefficient calculations were polynomial identities, so parameter values that remove further edges require no separate genericity argument. $\square$

Example 5.2 (A nonconstant generating determinant). Take $p(X) = X^2 - X + 2$, $a = 2/3$, and $q(X) = X^2 + X + 1$. Then (5.2) becomes

$$D(z)^{-1} = (1 - z)(1 - 2z/3)(1 - 2z - 2z^2/3).$$

The first five residue coefficients are

$$\rho_1 = \frac{11}{3}, \quad \rho_2 = \frac{61}{9}, \quad \rho_3 = \frac{359}{27}, \quad \rho_4 = \frac{2329}{81}, \quad \rho_5 = \frac{15611}{243}.$$

These are obtained from the three blocks in (5.6), or equivalently from the logarithm of the displayed polynomial. The signed stability trace, when interpreted pointwise, is their negative. Thus the uniform theorem contains cases beyond the low-degree vanishing of Remark 4.2.

Example 5.3 (A nonreduced fixed point). Let $p(X) = X^2 + 1$, $a = 1$, and $q(X) = X^2 + X + 1$. The period-one equation is $(X - 1)^2 = 0$. At the fixed point $(1, 1)$, $\det(I - DH) = 0$, so the ordinary pointwise summand is undefined. The residue convention instead gives

$$\rho_1 = \text{Res}_{X=1} \frac{(X^2 + X + 1) dX}{(X - 1)^2} = q'(1) = 3, \quad \tau_1 = -3.$$

For all periods, Proposition 5.1 gives

$$D(z)^{-1} = (1 - z)^2(1 - z - z^2).$$

Its residue coefficients begin 3, 5, 6, 9. These values include the degenerate scheme. They are not values assigned to the undefined pointwise quotient by cancelling or removing its denominator.

5.2 A complete ordinary trace-class boundary

The finite-state residue formula does not itself supply an ordinary operator with traces τ_n . In particular, replacing W by $-W$ would give $(-1)^n \text{Tr}(W^n)$, not $-\text{Tr}(W^n)$ for every n . There is a complete criterion for the existence of any ordinary trace-class representative of the frozen sequence.

Proposition 5.4. *For fixed parameters (2.1), a trace-class operator T on some complex Hilbert space satisfies*

$$\text{Tr}(T^n) = \tau_n \quad \text{for every } n \geq 1 \quad (5.7)$$

if and only if W is nilpotent.

Proof. If T is trace class, its ordinary Fredholm determinant $D_T(z) = \det(I - zT)$ is entire, and near zero it has the trace expansion

$$D_T(z) = \exp \left(- \sum_{n \geq 1} \frac{\text{Tr}(T^n)}{n} z^n \right).$$

These are the classical determinant properties summarized in Bornemann [1, Sec. 3, in particular Eq. (3.3) of the author manuscript]. By (5.7) and Theorem 2.2,

$$D_T(z) \det(I - zW) = 1$$

near zero. Both sides are entire functions, so the identity theorem extends the equality to all $z \in \mathbb{C}$. If $\det(I - zW)$ were nonconstant, it would have a complex zero, contradicting this identity. Hence $\det(I - zW) = 1$.

For a finite complex matrix, this last condition means that all eigenvalues are zero. Equivalently, its characteristic polynomial is a pure power of its variable, and the Cayley–Hamilton theorem implies that W is nilpotent. Conversely, if W is nilpotent then $\text{Tr}(W^n) = 0$ for every positive n . Thus all τ_n vanish, and $T = 0$ is a trace-class representative. $\square$

The criterion uses the classical obstruction that an ordinary Fredholm determinant is entire while a nonconstant reciprocal polynomial has a pole. It is a consequence of the specific residue sequence, not a separate construction of a dynamical operator. It does not exclude supertraces, distributional traces, different observables, or operators with absolute stability weights.

6 Scope and reproducibility

Theorem 2.2 replaces the period-dependent cyclic quotient algebras by a single edge-state matrix for a fixed map and coordinate weight. Its local bound is uniform in the period, but the matrix size still depends on the degree bound m . No minimal-dimension assertion is made. A different degree bound for the same polynomial can give a larger matrix with the same power traces and generating function.

The scope is tied to both the domain and the numerator. The residue includes every finite complex fixed point, with its scheme structure. Restricting to a real invariant set would change the sum. An absolute stability denominator is not the signed holomorphic denominator in (2.8). The argument also does not cover arbitrary two-variable weights, nonpolynomial weights, or compositions with a time-dependent polynomial. These changes are outside the theorem, irrespective of whether a finite representation might exist for some of them.

The quadratic calculation displays a loss of information in this observable: the coefficients c, w in (5.1) disappear from (5.2). The residue determinant therefore does not encode all parameters of the underlying map and weight. Its rationality, and its coefficient ring in (2.15), do not supply target Euler factors, root numbers, automorphy, a correspondence of zero divisors, or a Hilbert–Pólya realization.

Exact finite checks. The accompanying files `exact_check.py` and `EXACT_CHECK_OUTPUT.json` record exact symbolic checks performed during proof development. Across seven choices of p, q, a with $d = 2, 3, 4$, there are 27 comparisons through period five between the edge-matrix power traces and the normal-form coefficient in (3.1). Two nondegenerate fixed schemes are also checked by constructing quotient-algebra multiplication matrices and evaluating $\text{Tr}(m_{g_n} m_{\det J_n}^{-1})$. The checks assert that this inverse exists in those examples. The output includes six symbolic Hill identities, for $n = 1, \dots, 6$, and a direct symbolic calculation of the quadratic determinant with free coefficients.

The stored run used Python 3.12.3 and SymPy 1.14.0 and exited successfully. These checks test signs, local coefficients, short periods, and an explicit degeneracy. They were not rerun to produce the manuscript, and they do not establish its all-period quantifier. That quantifier follows from Lemma 4.1 and the bijection (4.10). No empirical data or stochastic numerical experiment is used in the proof.

Conclusion. The finite-state result follows from a local fact about every nonzero Laurent configuration: nearest-neighbour propagation and the factor $d > 1$ impose a period-independent exponent bound. This converts the complete global residue sum into the power trace of one polynomial matrix. The residue definition keeps the result valid at degenerate parameters, while the Hill sign fixes the reciprocal determinant and the ordinary trace-class boundary.

Availability and declarations

Data and code availability. The accompanying research package contains the complete proof, the symbolic checker, its saved exact output, and source-verification notes. The mathematical argument is contained in this manuscript. No empirical dataset is required.

Ethics. This mathematical work involves no human participants, animal experiments, or personal data.

Authorship, funding, and interests. This anonymous working draft does not identify human authors. CRediT assignments, funding information, and conflict-of-interest declarations have not been supplied and require author confirmation before any external submission.

AI assistance. The proof development, code, manuscript, and internal checking in the research package used AI assistance. The internal checks do not constitute external peer review. No external submission is asserted.

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